SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Technical Presentation

COURSE NAME: Data Structure **COURSE CODE: CSA03**
Assessment Tool Weightage: 30%

COURSE OUTCOME COVERED

CO1: *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL3, BL4)*

Data Structure Selection for Navigation Systems

Storing City Connections & Finding Optimal Shortest Routes

PRESENTER NAME	REGISTER NUMBER	COURSE DETAILS	ASSESSMENT
PAVITHRA R	192421369	Data Structure (CSA03)	Assessment Tool 2 (Weightage: 30%)

01 / PROBLEM STATEMENT

Designing an efficient backend data architecture for modern GPS navigation systems to store complex road networks and compute real-time shortest routes.

Selection: Weighted Directed Graph

Why Non-Linear Structure?

Road networks are inherently non-linear and interconnected. Cities and intersections branching into multiple roadways cannot be modeled sequentially without massive redundancy or loss of relational context.

Weighted Directed Graph (G = V, E)

A Weighted Directed Graph perfectly captures asymmetry (one-way streets, varying traffic speeds) and assigns real-world costs (distance, travel time, tolls) to edges between location nodes.

Graph Representation Mapping

Navigation Primitive	Graph Component	Attributes & Role
Cities / Intersections	Vertices (Nodes, V)	Stores location IDs, GPS coordinates (latitude/longitude), and landmark metadata.
Roads / Highways	Edges (Links, E)	Defines traversable paths connecting two physical location nodes.
Distance / Time / Toll	Edge Weights (w)	Numeric cost factor evaluated during route optimization algorithms.
One-Way / Restrictions	Directed Edges ($u \to v$)	Ensures movement is strictly constrained to legal traversal directions.

Storage Choice: Adjacency List

Evaluation Metric	Adjacency List CHOSEN	Adjacency Matrix	Impact on Navigation
Space Complexity	$O(\|V\| + \|E\|)$	$O(\|V\|^2)$	Saves massive memory since road networks are sparse ($\deg(v) \ll \|V\|$).
Neighbor Iteration	$O(\deg(v))$	$O(\|V\|)$	Fast retrieval of directly adjacent connecting roads.
Dynamic Updates	$O(1)$ Insertion	$O(\|V\|^2)$ Resize	Easily handles addition of new roads or traffic closures.

Shortest Route Pathfinding

Dijkstra's Algorithm

Computes single-source shortest path for non-negative edge weights using a Min-Priority Queue (Binary/Fibonacci Heap).

$$O((|V| + |E|) \log |V|)$$

A* Search Algorithm

Enhances Dijkstra by applying a heuristic function $h(v)$ (Euclidean distance to destination) to guide the directional search space.

$$f(n) = g(+ h(n)$$

Contraction Hierarchies

Speed-up technique pre-processing highway networks to achieve sub-millisecond query responses across continental route calculations.

Why Alternatives Are Unsuitable

- ✕ **Arrays & Linked Lists (Linear):** Fail due to $O(N)$ linear lookup times and inability to model complex multi-branching road networks without duplication.
- ✕ **Trees (Hierarchical):** Fail because trees strictly forbid cycles and enforce a single parent per node. Road networks contain alternative loops, cycles, and multi-path detours.
- ✕ **Adjacency Matrices:** Inefficient due to $O(|V|^2)$ memory waste; storing millions of cities would require terabytes of unneeded zero entries.

Performance & Complexity Summary

$$O(|V| + |E|)$$

Optimal Space Complexity

Linear space footprint relative to physical road count

Memory Efficiency

A network with 1,000,000 cities ($|V|$) and 3,000,000 roads ($|E|$) consumes less than 100 MB of RAM using an Adjacency List.

Query Speed

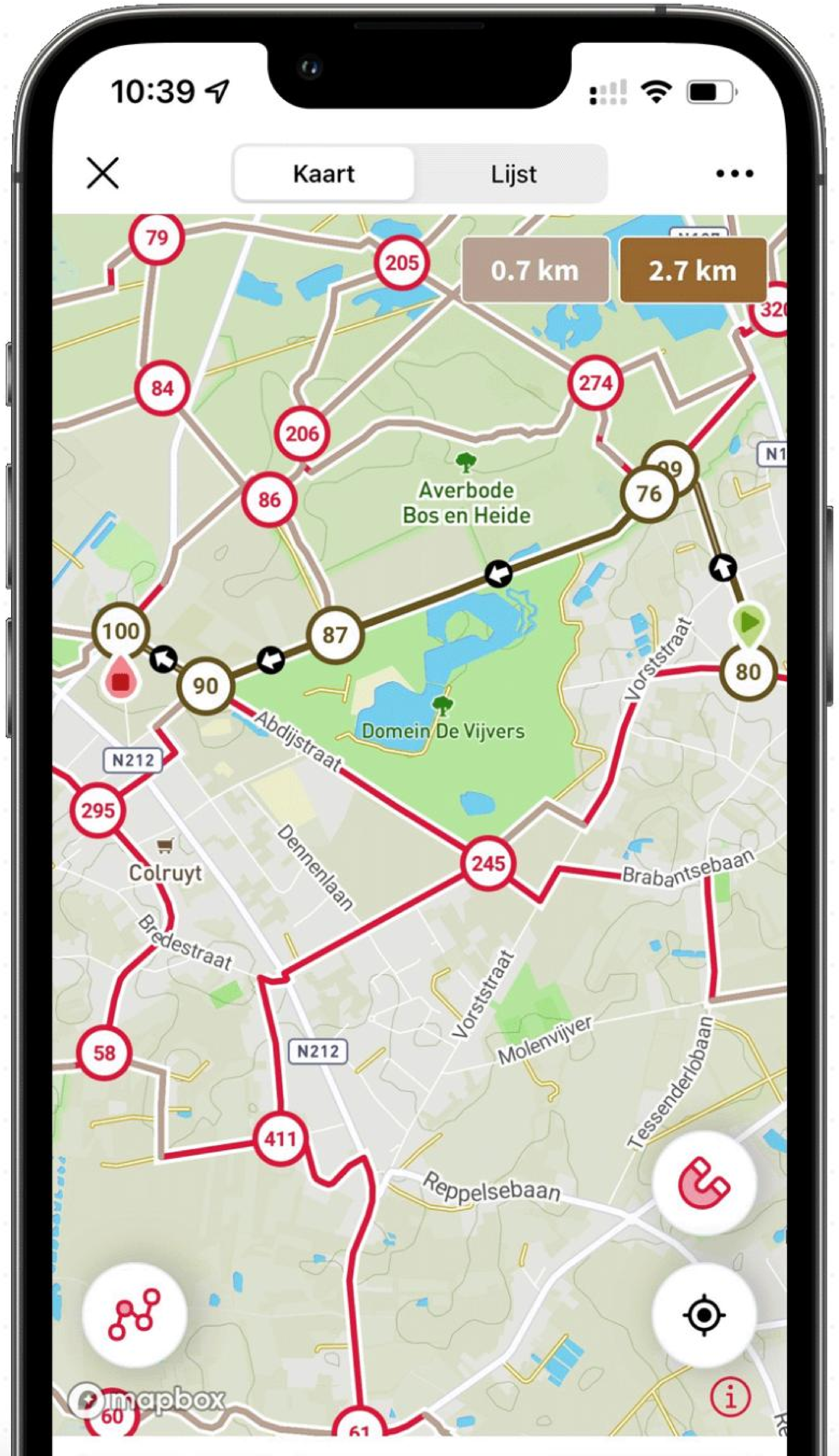
With Min-Heap priority queue optimization, shortest routes are computed in milliseconds for city-wide scale navigation.

Real-Time Updates

Dynamic Weight Re-calibration

When live traffic accidents or construction delays occur, the graph dynamically updates edge weights ($w_{new} = w_{base} \times \text{congestion}$) in $O(1)$ time without rebuilding the structure.

This agility allows navigation software to recalculate optimal detours mid-transit instantly.



Thank You!

Questions & Technical Discussion

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Image Sources

https://www.nodemapp.com/assets/img/custom/phone_2.png

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